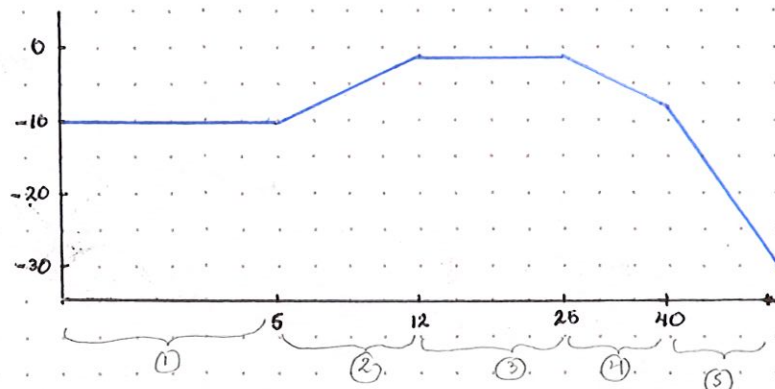


Por Anderson Imbaquingo

EVALUACIÓN 4

1.- La siguiente gráfica representa la magnitud de una función de transferencia de un circuito. Encuentre la función de red $H(j\omega)$



(1) Para $\omega \leq 5 \text{ rad/s}$

$$20 \log_{10}(K) = -10$$

$$\log_{10}(K) = -\frac{10}{20}$$

$$\log_{10}(K) = -\frac{1}{2}$$

$$K = e^{-\frac{1}{2}} \Rightarrow K = 0,316$$

(2) $\omega_1 = 5 \text{ rad/s}$ La pendiente cambia de 0 a +20 dB/década

$$0 + 20 = +20 \text{ dB/década} \quad \left. \begin{array}{l} \text{Pendiente} \\ \text{positiva} \end{array} \right\}$$

Pendiente Positiva $\Rightarrow$ Cero simple

$$\left(1 + \frac{s}{5}\right)$$

(3) Para $\omega_2 = 12 \text{ rad/s}$

$$20 \text{ dB/década} \Rightarrow 0 \text{ dB/década}$$

Pendiente negativa $\Rightarrow$ Polo Simple

$$\left(1 + \frac{s}{12}\right)$$

(4) $\omega_3 = 25 \text{ rad/s}$

$$0 \text{ dB/década} \Rightarrow -20 \text{ dB/década}$$

Pendiente negativa $\Rightarrow$ Polosimple

$$\left(1 + \frac{s}{25}\right)$$

Función de Transferencia

(5) $\omega_4 = 40 \text{ rad/s}$

$$-20 \text{ dB/década} \Rightarrow -40 \text{ dB/década}$$

Pendiente Negativa en -20 dB/década $\Rightarrow$ Polo Simple

$$\left(1 + \frac{s}{40}\right)$$

$$H(s) = K \frac{\text{Ceros}}{\text{Polos}}$$

$$H(s) = 0,316 \frac{\left(1 + \frac{s}{5}\right)}{\left(1 + \frac{s}{12}\right) \left(1 + \frac{s}{25}\right) \left(1 + \frac{s}{40}\right)}$$

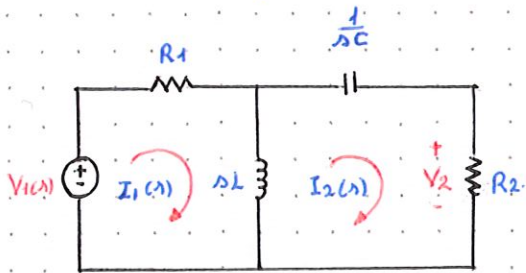
$$H(j\omega) = 0,316 \frac{\left(1 + \frac{j\omega}{5}\right)}{\left(1 + \frac{j\omega}{12}\right) \left(1 + \frac{j\omega}{25}\right) \left(1 + \frac{j\omega}{40}\right)}$$

$$\left(1 + \frac{j\omega}{12}\right) \left(1 + \frac{j\omega}{25}\right) \left(1 + \frac{j\omega}{40}\right)$$

2.- Para el siguiente circuito, determine:

a.- la función de transferencia $H(s) = \frac{I_2(s)}{V_1(s)}$

b.- El diagrama de Bode (magnitud y fase). Para este ítem asuma $R_1 = R_2 = 10 \text{ } \Omega$, $C = 2 \text{ F}$, $L = 2 \text{ H}$.



a) Malla ①

$$V_1(s) = I_1(s) R_1 + (I_1(s) - I_2(s)) \omega L$$

$$(R_1 + \omega L) I_1(s) - \omega L I_2(s) = V_1(s)$$

Malla ②

$$0 = (I_2(s) - I_1(s)) \omega L + I_2(s) \frac{1}{\omega C} + I_2(s) R_2$$

$$-\omega L I_1(s) + \left(\omega L + \frac{1}{\omega C} + R_2 \right) I_2(s) = 0$$

$$H(s) = \frac{I_2(s)}{V_1(s)} \quad \begin{cases} (R_1 + \omega L) I_1(s) - \omega L I_2(s) = V_1(s) \\ -\omega L I_1(s) + \left(\omega L + \frac{1}{\omega C} + R_2 \right) I_2(s) = 0 \end{cases}$$

$$\begin{bmatrix} R_1 + \omega L & -\omega L \\ -\omega L & \omega L + R_2 + \frac{1}{\omega C} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \end{bmatrix}$$

$$\Delta = (R_1 + \omega L) \left(\omega L + R_2 + \frac{1}{\omega C} \right) - (-\omega L)(-\omega L)$$

$$\Delta = (R_1 + \omega L)(\omega L) + (R_1 + \omega L)R_2 + (R_1 + \omega L)\frac{1}{\omega C} - \omega^2 L^2$$

$$\Delta = R_1 \omega L + \cancel{\omega^2 L^2} + R_1 R_2 + \omega L R_2 + \frac{R_1}{\omega C} + \frac{\omega L}{\omega C} - \omega^2 L^2$$

$$\Delta = \omega L (R_1 + R_2) + R_1 R_2 + \frac{L}{C} + \frac{R_1}{\omega C}$$

$$\Delta = \frac{\omega^2 L C (R_1 + R_2) + \omega C (R_1 R_2) + \omega L + R_1}{\omega C}$$

$$\Delta = \frac{\omega^2 L C (R_1 + R_2) + \omega (R_1 R_2 C + L) + R_1}{\omega C} \neq$$

$$\Delta_2 = \begin{bmatrix} R_1 + \omega L & V_1 \\ -\omega L & 0 \end{bmatrix}$$

$\Rightarrow$ Función de Transferencia

$$\frac{I_2(s)}{V_1(s)} = \frac{(\omega L)(\omega C)}{\omega^2 L C (R_1 + R_2) + \omega (R_1 R_2 C + L) + R_1}$$

$$H(s) = \frac{\omega^2 L C}{\omega^2 L C (R_1 + R_2) + \omega (L + R_1 R_2 C) + R_1}$$

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$I_2 = \frac{\omega L V_1}{\omega^2 L C (R_1 + R_2) + \omega (R_1 R_2 C + L) + R_1} \neq$$

$$(b) H(s) = \frac{s^2 LC}{s^2 LC (R_1 + R_2) + s(L + R_1 R_2 C) + R_1} = \frac{s^2 (2)(2)}{s^2 (2)(2)(10+10) + s(2 + (10)(10)(2)) + 10}$$

$$H(s) = \frac{4s^2}{80s^2 + 202s + 10} = \frac{2s^2}{40s^2 + 101s + 5}$$

$$\Rightarrow 40s^2 + 101s + 5 \Rightarrow s = \frac{-101 \pm \sqrt{101^2 - 4(40)(5)}}{2(40)} \Rightarrow s_1 = \frac{-11}{80} = -0,05$$

$$s_2 = \frac{-198}{80} = -2,5$$

Pólos

$$H(s) = \frac{2s^2}{40(s + 0,05)(s + 2,5)} = \frac{2s^2}{40[0,05(1 + \frac{s}{0,05})][2,5(1 + \frac{s}{2,5})]}$$

$$H(s) = \frac{2s^2}{5(1 + \frac{s}{0,05})(1 + \frac{s}{2,5})} = \frac{0,4}{(1 + \frac{s}{0,05})(1 + \frac{s}{2,5})} \neq$$

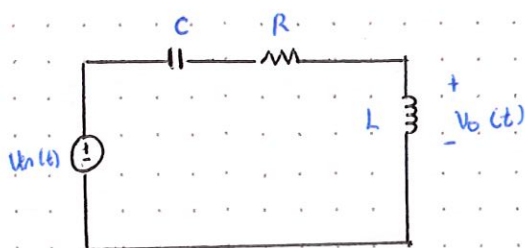
$$K = 0,4 \Rightarrow 20 \log(0,4) = -8 \text{ dB}$$

3.- Dado el siguiente circuito. Determine:

a.- Obtenga la función de transferencia $\frac{V_o(j\omega)}{V_{in}(j\omega)}$

b.- Grafique el diagrama de bode (magnitud), ¿Qué tipo de filtro es? Para este ítem asuma $C = 0,5 \text{ F}$, $R = 1 \Omega$ y $L = 2 \text{ H}$

c.- Obtenga el valor de R , si se conoce que la frecuencia de corte (ω_c) es de $0,1 \text{ rad/s}$. $C = 0,1 \text{ F}$ y $L = 1 \text{ H}$.



a) $\rightarrow$ Divisor de Voltaje.

$$H(s) = \frac{Z_L}{Z_R + Z_C + Z_L}$$

$$H(s) = \frac{sL}{R + \frac{1}{sC} + sL}$$

$$H(s) = \frac{s^2 L}{sR + \frac{1}{C} + s^2 L}$$

$$H(s) = \frac{s^2}{s^2 + s \frac{R}{L} + \frac{1}{LC}} \neq$$

b) $R = 1 \Omega$, $L = 2 H$, $C = 0,5 F$

$$H(s) = \frac{s^2}{s^2 + s \frac{R}{L} + \frac{1}{LC}} = \frac{s^2}{s^2 + \frac{1}{2} + 1} \quad \neq$$

$\Rightarrow$ Comprobación de filtro

$\omega \rightarrow 0 : |H| \rightarrow 0 \Rightarrow$ Pasa Bajos

$\omega \rightarrow \infty : |H| \rightarrow 1 \Rightarrow$ Pasa Alto

$$\lim_{s \rightarrow \infty} \frac{s^2}{s^2} = 1 \quad \neq$$

Con $|H| \rightarrow 1$ es un filtro Pasa Alto $\neq$

c) $\omega_c = 10 \text{ rad/s}$, $C = 0,1 F$ y $L = 1 H$

$$H(s) = \frac{s^2}{s^2 + s \frac{R}{1} + \frac{1}{(1)(0,1)}} = \frac{s^2}{s^2 + sR + 10}$$

$$H(j\omega) = \frac{(j\omega)^2}{(j\omega)^2 + (j\omega)R + 10}$$

$\omega_c \Rightarrow 10 \text{ rad/s}$

$$H(j\omega) = \frac{(j(10))^2}{(j(10))^2 + (j(10))R + 10} = \frac{-100}{-100 + j10R + 10} = \frac{-100}{-90 + j10R}$$

$\bullet 20 \log_{10}(K) = -3 \text{ dB}$

$|H(j\omega)| = 0,707$

$\log_{10}(K) = \frac{-3}{20}$

$K = e^{-3/20}$

$K = 0,707 \text{ dB} \quad \neq$

$$|H(j\omega)| = \frac{|-100|}{\sqrt{(-90)^2 + (10R)^2}} = \frac{100}{\sqrt{8100 + 100R^2}}$$

$$\frac{100}{\sqrt{8100 + 100R^2}} = 0,707$$

$$(\sqrt{8100 + 100R^2})^2 = \left(\frac{100}{0,707}\right)^2$$

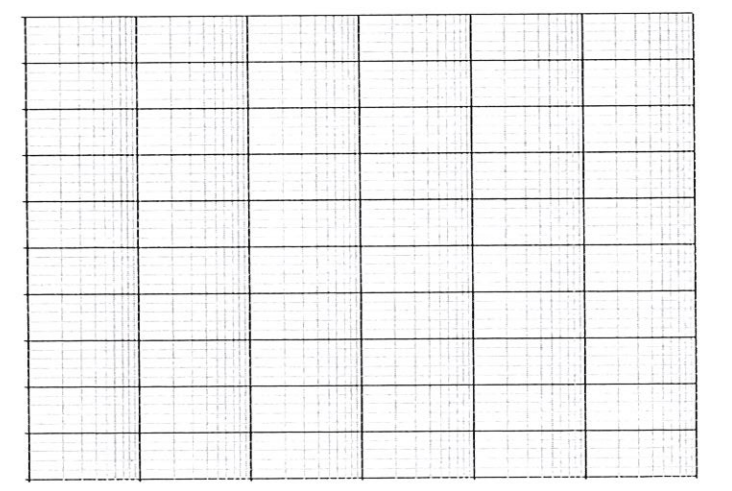
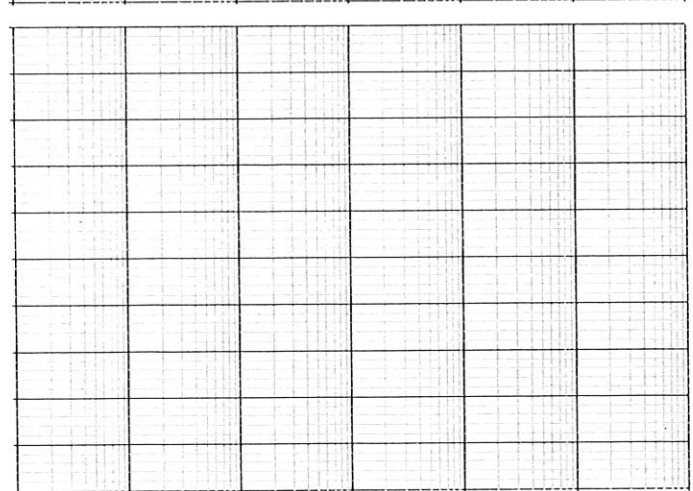
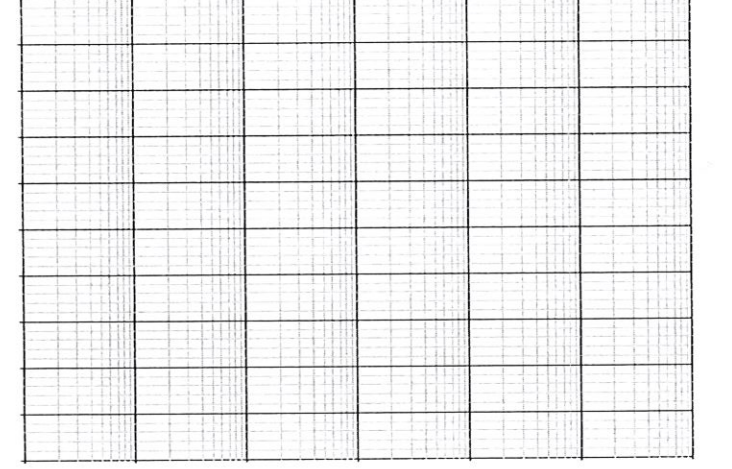
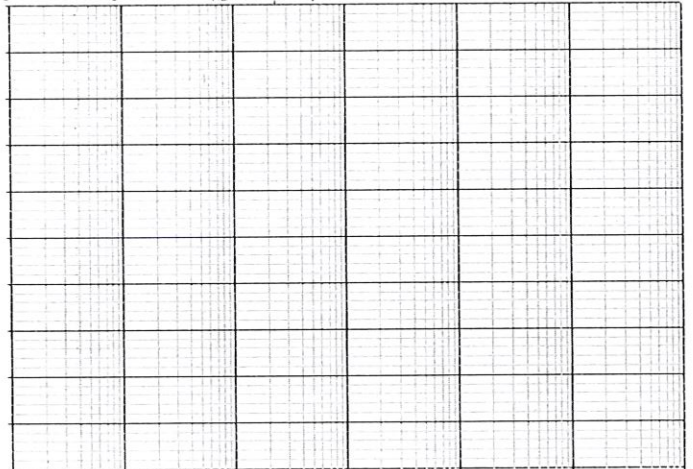
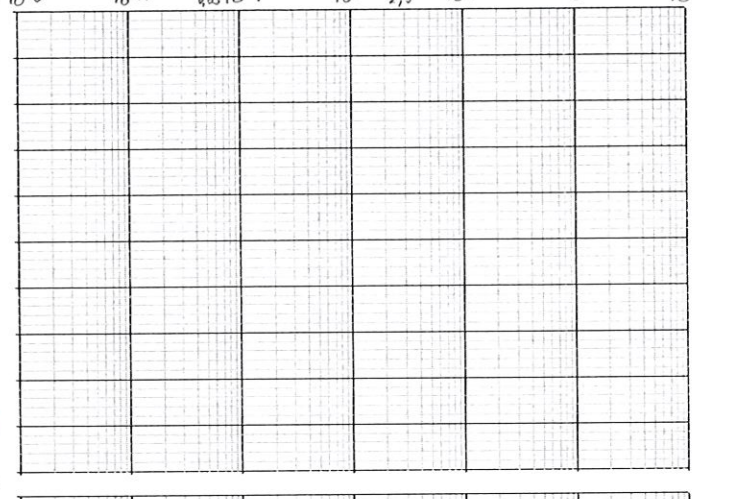
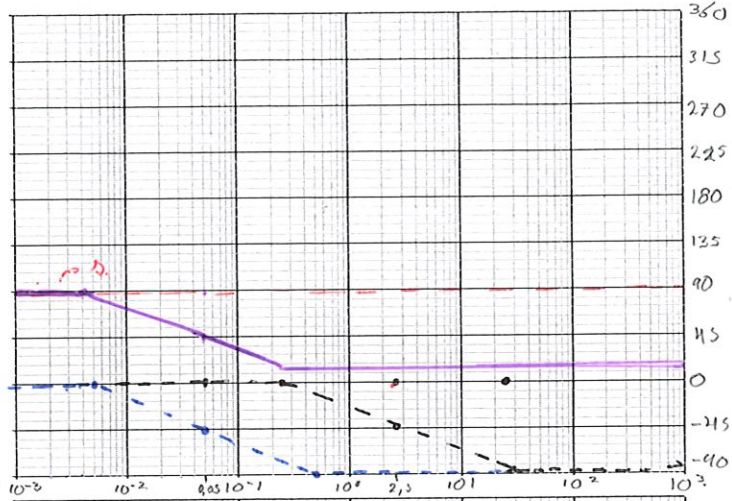
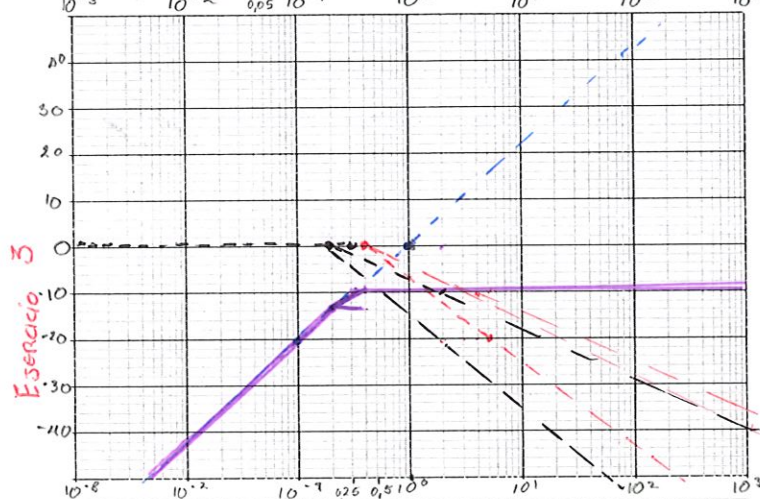
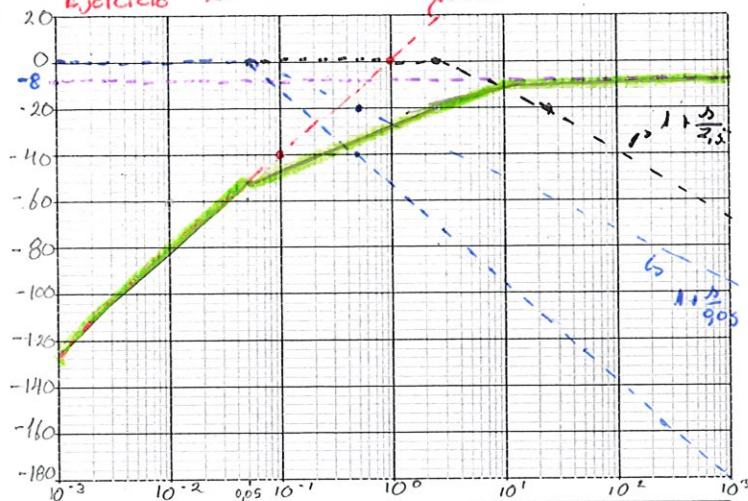
$$100R^2 = \left(\frac{100}{0,707}\right)^2 - 8100$$

$$R = \sqrt{\frac{\left(\frac{100}{0,707}\right)^2 - 8100}{100}}$$

$R = 10,91 [\Omega] \quad \neq$

Ejercicio 2.

G. H.



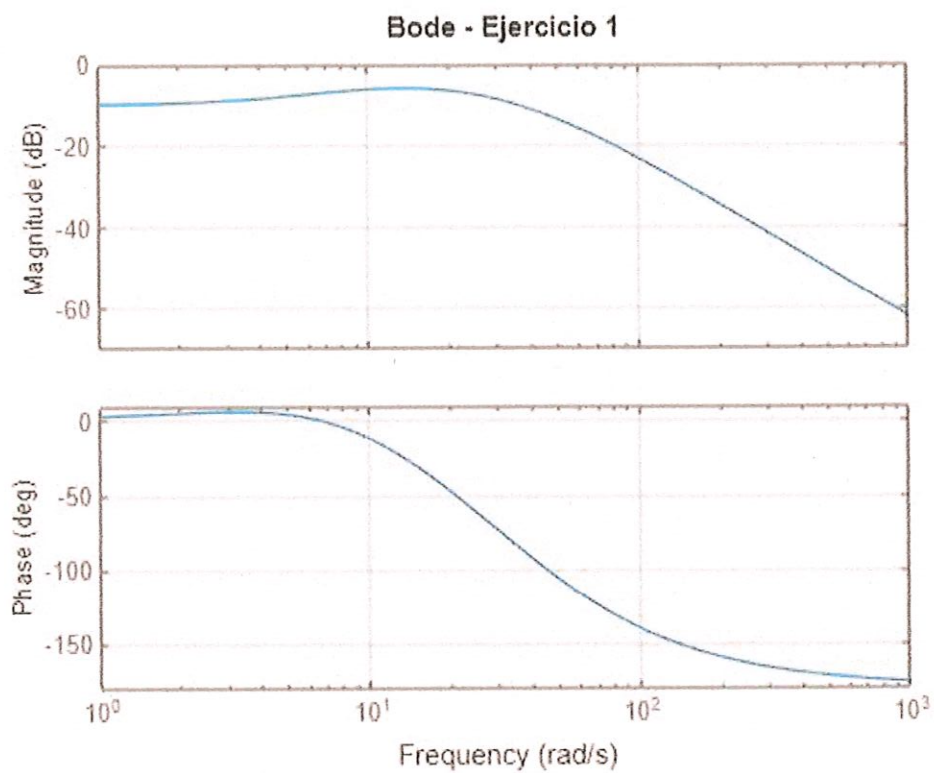
ESCUELA POLITÉCNICA NACIONAL

ANÁLISIS DE CIRCUITOS ELÉCTRICOS

PRUEBA 4

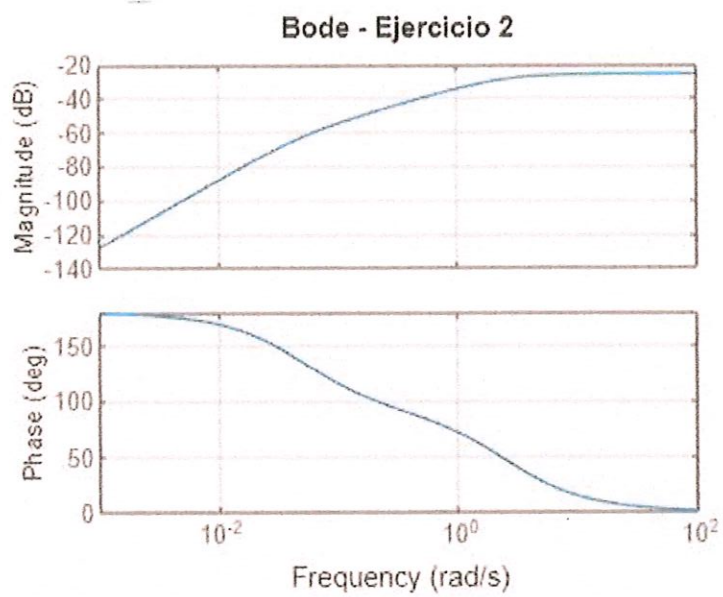
EJERCICIO 1

```
clc; clear; close all;  
  
s = tf('s');  
  
H1 = 0.316*(1 + s/5)/((1 + s/12)*(1 + s/25)*(1 + s/40));  
  
figure;  
bode(H1)  
grid on  
title('Bode - Ejercicio 1')
```



EJERCICIO 2

```
clc; clear; close all;  
  
s = tf('s');  
  
H2 = (2*s^2)/(40*s^2 + 101*s + 5);  
  
figure;  
bode(H2)  
grid on  
title('Bode - Ejercicio 2')
```



EJERCICIO 3

```
clc; clear; close all;  
  
s = tf('s');  
  
H3 = (s^2)/(s^2 + s/2 + 1);  
  
figure;  
bode(H3)  
grid on  
title('Bode - Ejercicio 3')
```

